TT-E-001384(GSA-FSS)
August 5, 1979
SUPERSEDING
(see 6.4)

INTERIM FEDERAL SPECIFICATION

ENAMEL, GLOSS, SYNTHETIC (FOR METAL EQUIPMENT)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two classes of fast drying enamel suitable primarily for application to metal equipment.

1.2 Classification. The enamel shall be of the following classes as specified in the invitation for bids, contract or purchase order (see 6.3).

Class 1 - Conventional ready-mixed form, in bulk containers, suitable primarily for spray application.
Class 2 - Enamel in pressurized dispensers for touch-up of areas to which the class 1 enamel has been applied.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-P-1385 - Primer Coating, Synthetic, Gray (For Metal Equipment).
TT-T-1386 - Thinner, Synthetic Enamel (for Low and High Temperature Spray Application).
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO: Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Code of Federal Regulations:

49 CFR 171-178 - Department of Transportation Regulations for the Transportation of Explosives and Other Dangerous Articles by Land and Water.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 523 - Specular Gloss.
D 562 - Consistency of Paints Using the Stormer Viscosimeter.
D 609 - Preparation of Steel Panels for Testing Paint, Varnish, Lacquer, and Related Products.
D 822 - Light- and Water-Exposure Apparatus (Carbon-Arc Type) for Testing Paint, Varnish, Lacquer, and Related Products on Test Panels.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems.
D 1640 - Drying, Curing, or Film Formation of Organic Coatings at Room Temperature.
D 1644 - Nonvolatile Content of Varnishes.
D 1729 - Visual Evaluation of Color Differences of Opaque Materials.
D 2244 - Instrumental Evaluation of Color Difference of Opaque Materials.
D 2698 - Pigment Content of Solvent-Type Paints by High Speed Centrifuging.
D 2805 - Hiding Power of Paints.
D 3272 - Vacuum Distillation of Solvents from Solvent Base Paints for Analysis.
E 260 - General Gas Chromatography procedures.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Materials.

3.1.1 Solvent (class 1 only). The volatile solvent, when tested as specified in 4.4.1, shall conform to the following requirements by volume:

(a) A total of solvents with olefinic or cycloolefinic unsaturation shall not exceed 5 percent.
(b) The total aromatic compounds with eight or more carbon atoms in the molecule, except ethylbenzene, methyl benzoate, and phenyl acetate, shall not exceed 8 percent.
(c) The total of ethylbenzene, toluene, and branched-chain ketones shall not exceed 20 percent.
(d) A solvent which may be classified into more than one of the above groups shall be considered a member of the group having lowest allowable concentration.
(e) The total of (a) + (b) + (c) shall not exceed 20 percent.

3.1.2 Vehicle. When the nonvolatile vehicle is tested as specified in 4.4.7, band positions, shapes, and relative intensities in each spectrum shall match the spectrum in figure 1.

3.1.3 Propellant (class 2). The propellant shall be a hydrocarbon or a blend of hydrocarbons.

3.2 Specific requirements. Each color shall meet the requirements specified in table I and table II when tested as specified in 4.4.

3.3 Pressurized dispenser (class 2).

3.3.1 Container. The container shall be a nominal 16-ounce commercial type metal pressure container generally known as an aerosol container of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with generally recognized safe commercial practice. Each container shall have a removable, press-fitting metal or plastic cover cap. Each container shall be labeled as stated in section 5.

3.3.2 Spray delivery. When tested as specified in 4.4.18.2.1, the contents of the dispenser container shall be sprayable when stabilized at temperatures from 16° to 32°C. A minimum of 98 percent by weight of content shall be sprayable in a pattern such as to deposit a uniform coating on a smooth vertical surface. The film shall dry free of orange peel, sags, specks, seeds, or any other surface irregularities. Leftover enamel of more than 2 percent shall constitute a defect.

3.4 Qualitative requirements.

3.4.1 Condition in container (class 1 only). A freshly opened full container of the enamel, when tested as in 4.4.15, shall be free from grit, seeds, skins, lumps, abnormal thickening, or livering, and shall show no more pigment settline or caking than can be readily reincorporated to a smooth homogeneous state.

3.4.2 Storage stability (class 1 only). A partially filled container of the enamel shall show no skinning when tested as specified in 4.4.16.

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TABLE I

Nonvolatile, pigment, and contrast ratios

Class 1 enamel					Class 2 enamel				
Color and name (Fed. Std. No. 595)	Nonvolatile min. (% by wt. of enamel)	Nonvolatile vehicle, min. (% by wt. of vehicle)	Pigment, min. (% by wt. enamel)	Contrast ratio, min.	Color and name (Fed. Std. 595)	Total nonvolatile in dispenser, min. (grams)	Pigment min. (% by wt. of total nonvolatile)	Nonvolatile vehicle, min. (% by wt. of total nonvolatile)	Contrast ratio, min.
11136, red	50.0	44.0	10.0	0.87	11136 red	47.0	21.9	77.0	0.72
14064, olive green	45.0	43.0	3.5	0.94	14064, olive green	51.5	7.0	92.0	0.99
15050, blue	45.0	43.0	3.7	0.96	15050, blue	47.5	9.3	89.7	0.96
17773, white	50.0	43.0	17.0	0.96	17773, white	81.0	23.0	76.0	0.90

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TABLE II. Physical and chemical characteristics

Characteristics	Requirements			
	Class 1		Class 2	
	Min.	Max.	Min.	Max.
Fineness of grind	7	----	7	----
Consistency (KU; shearing rate 200 r.p.m.)	60	72	----	----
Drying time:				
Set to touch (minutes)	----	20	----	20
Dry hard (hours)	----	24	----	24
Dry for recoat (hours)	----	24	----	24
Dry for taping (hours)	----	3	----	3
Accelerated weathering for 300 hours:				
60° specular gloss				
Before weathering	90	----	90	----
After weathering	80	----	80	----
Color difference (ΔE)				
After weathering	----	2.0	----	2.0
Chalking				
After weathering	none	none	none	none
Hardness (Sward)				
After drying 1 week	10	----	10	----
Net contents, by weight, of dispenser (grams)	----	----	360	----

3.4.3 Compatibility with thinner (class 1 only). When tested as specified in 4.4.17, the enamel shall show no evidence of coagulation, separation, precipitation, or incompatibility with the thinner.

3.4.4 Spraying properties (class 1 only). When tested as specified in 4.4.18.1, the enamel shall spray satisfactorily without running, sagging, or streaking. The dried film shall show no dusting, mottling, or color separation, and shall have a smooth, uniform finish free from seeds.

3.4.5 Color. The enamel shall match the specified color in Fed. Std. No. 595 (see 3.2.1), when tested as specified in 4.4.10.

3.4.6 Flexibility. When tested as specified in 4.4.12, a dry film of the enamel shall withstand bending without cracking or flaking.

3.4.7 Resistance to water immersion. When tested as specified in 4.4.13, a dry film of the enamel shall resist degradation when immersed in distilled water.

3.4.8 Lifting (class 2 only). When tested as specified in 4.4.18.2.2, the enamel shall not lift a dry film of itself, or a dry film of the class 1 enamel. When tested in a similar manner, a dry film of the class 2 enamel shall not be lifted by the class 1 enamel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Inspection of preparation for delivery.

4.2.1 Class 1. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

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4.2.2 Class 2. The packaging, packing, and marking shall be inspected for conformance with the requirements of 3.3, 5.2.2, and 5.3 as applicable. Inspection lots sampling shall be in accordance with MIL-STD-105, inspection level I. Defects indicating non-conformance with 3.3, 5.2.2, and 5.3 shall be inspected using acceptable quality level (AQL) of 1.5 percent defective. Defects indicating inadequate warning labeling (see 5.6 (g), (h), (i), and (j)) shall be inspected using an AQL of 0.015 percent defective.

4.3 Inspection of the end item.

4.3.1 Class 1. For end item inspection not involving tests, the filled unit containers of the enamel shall be inspected in accordance with MIL-STD-105, inspection level S-2, with an AQL of 4.0 percent defective.

4.3.2 Class 2. For end item inspection not involving tests, the filled pressurized dispensers shall be inspected in accordance with MIL-STD-105, level S-2, with an AQL of 4.0 percent defective.

4.3.3 Sampling of the end item for tests.

4.3.3.1 Class 1. For purposes of sampling, a lot shall be expressed in units of gallons. Samples from lots shall be taken in accordance with MIL-STD-105, using inspection level S-2 and an AQL of 2.5 percent defective.

4.3.3.2 Class 2. For purposes of sampling, a lot shall be expressed in units of filled pressurized dispensers. Samples from lots shall be taken in accordance with MIL-STD-105, using inspection level I and an AQL of 1.5 percent defective.

4.4 Test procedures. The enamel shall be tested in accordance with table III, and as hereinafter specified, to determine compliance with the applicable requirements in section 3. Unless otherwise specified, all tests shall be conducted at standard conditions, which are 23° ± 1°C and 50 ± 5 percent relative humidity. All test reports shall contain the individual values utilized in expressing the final result. Failure of any test, or noncompliance with any requirement shall be cause for rejection of the sample.

TABLE III. Index of tests

Test	ASTM Method	Test ref.	Rqmt. ref.
Nonvolatile content	D 1644	4.4.4	Table I
Nonvolatile vehicle	D 2698	4.4.5	Table I
Pigment	D 2698	4.4.6	Table I
Fineness of grind	D 1210	---	Table II
Consistency	D 562	---	Table II
Drying time	D 1640	4.4.8	Table II
Dry for taping	---	4.4.8.3	Table II
Contrast ratio	D 2805	4.4.9	Table I
Specular gloss	D 523	4.4.11	Table II
Accelerated weathering	D 822	4.4.11	Table II
Color	D 1729	4.4.10	3.4.5
Color difference	D 2244	4.4.11	Table II
Chalking	D 822	4.4.11	Table II
Hardness	---	4.4.14	Table II
Preparation of steel panels	D 609	4.4.18.2.2	3.3.5.3
Preparation of tin panels	2012 1/	4.4.11	Table II
		4.4.12	Table II
		4.4.13	Table II
Flexibility	6223 1/	4.4.12	3.4.6
Spraying properties	4331 1/	4.4.18.1	3.4.4
Lifting	---	4.4.18.2.2	3.4.8
Spray delivery	---	4.4.18.2.1	3.3.2
Condition in container	3011 1/	4.4.15	3.4.1
Compatibility with thinner	---	4.4.17	3.4.3
Immersion resistance	---	4.4.13	3.4.7
Infrared spectrophotometry	---	4.4.7	3.1.2
Solvent distillation	D 3272	4.4.1	3.1.1
Solvent analysis	D 260	4.4.1	3.1.1
Net contents	---	4.4.3	Table II

1/ Federal Test Method Std. No. 141.

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4.4.1 Solvent analysis (class 1). Analyze the solvent by gas chromatography. Separate the solvent from the nonvolatile as specified in ASTM D 3272, or inject the enamel directly into the gas chromatograph by equipment specifically designed for that purpose. The procedure specified in ASTM D 260, and any apparatus, operating conditions columns, and options permitted therein shall be used for the chromatographic analysis. The accuracy of the analysis shall be 0.25 percent, or less, for each component and the reproducibility shall be 0.25 percent absolute by weight over three or more runs. All peaks 0.5 percent or more of the sample shall be identified and quantified. Convert percent by weight to percent by volume, and evaluate results for compliance with 3.1.1.

4.4.2 Test concentrate (class 2). Tests on the class 2 enamel shall be conducted on a test concentrate prepared as follows: With the container well agitated and in an upright position, pierce the top with a very fine hole (with a needle point), which will allow the propellant to escape (conditioning at low temperature for 1 hour will facilitate this). After a few minutes, the pierced hole can be enlarged, and as soon as there is no further evidence of gas escaping, the top of the container can be cut open. Allow to stand until the contents are at room temperature then transfer the contents quantitatively into a tared beaker using several washes of acetone. Evaporate on a water bath until no more propellant is liberated and until the contents are less than 40 percent of the original weight of the contents including propellant. Then readjust to exactly 40 percent with n-butyl acetate.

4.4.2.1 For the qualitative tests, the manufacturer may use the enamel concentrate as it exists before packaging and pressurization, provided it is adjusted as specified in 4.4.2.

4.4.3 Total net contents (class 2). The total net contents shall be determined by calculating the difference in weight between the filled container and the completely empty one (see 4.4.2 for procedure of emptying). Wash out interior of can with acetone and dry. (Note that the weight of the agitators must be included in the tare.) Using the completely emptied container as tare, additional checks can be made on filled containers. At least two containers shall be checked from each lot sampled since variations in weight of empty cans can be found. A check of full and empty cans shall be made on the same cans.

4.4.4 Nonvolatile content.

4.4.4.1 Class 1. Test for nonvolatile content of the enamel, as received, in accordance with ASTM method D 1644.

4.4.4.2 Class 2. Use ASTM method D 1644 on the test concentrate. The nonvolatile figure obtained is multiplied by 0.040 to obtain the total solids in the pressurized container as received.

4.4.5 Nonvolatile vehicle.

4.4.5.1 Class 1. Test for nonvolatile vehicle in accordance with ASTM method D 2698 using the following mixture for extraction:

(a) 1 volume petroleum ether.
(b) 1 volume toluene.
(c) 1 volume acetone.
(d) 3 volumes absolute ethyl alcohol.

4.4.5.2 Class 2. Test for nonvolatile vehicle from the test concentrate described in 4.4.2, by ASTM method D 2698 using the following mixture for extraction:

(a) 1 volume petroleum ether.
(b) 1 volume toluene.
(c) 1 volume acetone.
(d) 9 volumes hot absolute ethyl alcohol.

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4.4.6 Pigment.

4.4.6.1 Class 1. Test for pigment by ASTM method D2698 using the extraction mixture specified in 4.4.5.1. Calculate pigment on the basis of total weight of the enamel.

4.4.6.2 Class 2. Test for pigment on the test concentrate by ASTM method D 2698 using the extraction mixture described in 4.4.5.2. Calculate pigment on the total nonvolatile content obtained as in 4.4.4.2.

4.4.7 Vehicle. Test a film of the nonvolatile vehicle, obtained by separation of the enamel as specified in 4.4.5, and oven dried for 1 hour at 105°C, in an infrared spectrophotometer.

4.4.8 Drying time.

4.4.8.1 Class 1. Except for the dry-for-taping time, test for all drying times by ASTM method D 1640, applying the enamel to a wet film thickness of 0.038 mm.

4.4.8.2 Class 2. Prepare four panels for drying time by spraying from the dispenser on each panel a dry film of approximately 0.012 mm. Test two of the panels for drying times specified in table I in accordance with ASTM method D 1640. Allow the other two panels to dry for 24 hours, then spray on each a second coat of approximately 0.012 mm of the enamel. Test the second coat for drying times specified in table I, in accordance with ASTM method D 1640.

4.4.8.3 Dry for taping. Apply the enamel to a smooth glass panel in such a manner as to give a dry film thickness of 0.051 ± 0.005 mm (0.002 ± 0.0002 in). After the film has dried in a horizontal position for 3 hours, apply across the entire width of the film a piece of 25 mm wide masking tape. Roll the tape smoothly and uniformly with an 11 kg roller. Remove the tape with a smooth rapid pulling. Any removal or deformation of the enamel film shall constitute failure to pass the dry-for-taping test.

4.4.9 Contrast ratio.

4.4.9.1 Class 1. Test in accordance with ASTM method D 2805 using a maximum dry film thickness of 0.048 mm (0.0015 in) for red 11105, and a maximum dry film thickness of 0.025 mm (0.001 in) for the other specified colors.

4.4.9.2 Class 2. Apply a wet film 0.051 mm in thickness, of the test concentrate obtained in 4.4.2 to a black and white hiding power chart. Application, drying, and testing of the film shall be in accordance with ASTM method D 2805.

4.4.10 Color. Apply the enamel, at a wet film thickness of 0.076 mm, to a clean, smooth, white panel. Allow the film to dry for 24 hours in a horizontal position and determine the color in accordance with ASTM method D 1729.

4.4.11 Accelerated weathering. Using tin panels as specified in method 2012 of Federal Test Method Std. No. 141, prepare six panels by applying sufficient enamel to a dry film thickness of 0.051 ± 0.005 mm. Air dry the panels at standard conditions for 96 hours. Expose three of the panels to accelerated weathering for 300 hours in accordance with ASTM method D 822. Air dry the panels for 2 hours at standard conditions (see 4.4) and examine for chalking. Wash the weathered panels with running water and a thoroughly degreased lamb's wool pad. Dry the panels with clean cheesecloth, then air dry for 2 hours as above. Test the weathered and unweathered panels for gloss in accordance with ASTM method D 523. Test the weathered and unweathered panels for color difference in accordance with ASTM method D 2244. Report values as the average for the triplicate panels.

4.4.12 Flexibility (cold cracking). The flexibility tests shall be as determined by method 6223 of Federal Method Std. No. 141. Test panels shall be as specified in method 2012 of the standard. Draw down films, to a dry thickness of 0.025 mm of the class 1 enamel, as received, or the test concentrate (see 4.4.2) of the class 2 enamel, air dry for 48 hours at standard conditions (see 4.4). The film shall withstand the specified cold cracking test when bent over a 6 mm rod.

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4.4.13 Immersion resistance. Using a film applicator with 0.152 mm gap clearance, apply films of the as received class 1 enamel, or test concentrate (see 4.4.2) of class 2 enamel, on tin panels as specified in method 2012 of Federal Test Method Std. No. 141. Air dry for 24 hours at standard conditions (see 4.4), then immerse for 16 hours in distilled water at 23° ± 1°C. Immediately upon removal from the water, the films shall show no wrinkling or blistering. After the panels are air dried for 24 hours, the films shall show no more than slight dulling or whitening.

4.4.14 Hardness. Apply films of the enamel of 0.076 mm in wet film thickness, to two smooth glass panels and air dry in a horizontal position for 1 week at standard conditions (see 4.4). Test the films for hardness with a Sward rocker in accordance with instructions of the manufacturer.

4.4.15 Condition in container (class 1 only). Determine condition in container by method 3011 of Federal Test Method Std. No. 141.

4.4.16 Storage stability (class 1 only). Test for skinning in a wide mouth glass jar approximately 111 mm in height and 51 mm in diameter, with a capacity of 240 ml, and having a tight fitting plastic screw top. Measure 180 ml of the well stirred enamel into the jar, and screw the top on securely. Invert the partially filled jar for 2 minutes, then turn the jar right side up and store undisturbed in a dark cabinet at standard conditions(see 4.4) for 48 hours. Without disturbing the contents, remove the screw top and examine the enamel for skinning.

4.4.17 Compatibility with thinner (class 1 only). Pour 100 ml of the well-mixed enamel into a 250-ml high-form glass beaker. Add 100 ml of thinner conforming to type I of TT-T-001386. Stir the enamel and thinner thoroughly but without making bubbles. Immediately flow out a thin film of the mixture on a clear glass plate. Examine the film for nonuniformities such as small gelled particles, which will indicate coagulation of the enamel vehicle and constitute failure of the requirements in 3.4.3. Small gelled particles may be differentiated from possible pigment flocculation by probing with a sharp-pointed metal object. After flowing out the films, allow the mixture in the beaker to remain undisturbed for 2 hours, and then examine the settled portion for possible coagulation of the enamel vehicle. Any settling or lumping which is not easily redispersed by light stirring with a glass rod shall constitute failure to meet 3.4.3.

4.4.18 Spraying properties.

4.4.18.1 Class 1. Reduce the enamel if necessary to spraying consistency according to the manufacturer's instructions using a thinner conforming to type I of TT-T-001386. Using any suitable, conventional, commercially available spray equipment, spray the enamel of proper consistency on a prepared wallboard panel. While spraying, hold the spray gun perpendicular to the panel and move it in straight lines back and forth across the panel, maintaining a distance of 8 to 10 inches between spray head and panel. Place the spray panel in a vertical position and observe the spray pattern for uniformity. Allow the sprayed film to dry, and examine it in accordance with method 4331 of Federal Test Method Std. No. 141 for compliance with 3.4.4.

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5.3 Palletization. When shipments to Government depots are full car or truckload, the shipping containers shall be palletized on expendable wooden pallets, 2 or 4 way design, to facilitate handling, in accordance with normal commercial practice. The palletized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be as specified in the contract or order (see 6.2).

5.5 DOT regulations. The packaging, packing, marking, and labeling shall comply with 49 CFR 171-178.

5.6 Special marking. Legible commercial marking shall be acceptable. Cans overcoated to obliterate original marking shall not be acceptable. Marking shall be lithographed on the unit container or printed on a lable securely affixed to the container and shall include the following:

(a) Name of material.
(b) Volume in units of US gallons (class 1).
(c) Net weight in ounces (class 2).
(d) Color number of Fed. Std. No. 595.
(e) Date of manufacture, in month and year.
(f) Date of first reinspection (Insert date one year after date of manufacture.)
(g) "Do not puncture or incinerate." (class 2 only).
(h) "Keep away from fire or flame." (class 2 only).
(i) "Use with adequate ventilation."
(j) "Store below 49°C (120"F). (class 2 only).

6. NOTES

6.1 Intended use. This enamel is used where there is need for a relatively fast drying, high gloss coating applied to automotive equipment, postal letter boxes, and other metal surfaces subject to sustained exterior exposure. The conventional ready mixed class 1 enamel may be brushed, but it is suited more to spray applications. The class 2 enamel is intended primarily for touch-up of the class 1 enamel.

6.1.1 Coating system. This enamel is intended as part of a coating system where the surface is first primed with primer coating conforming to TT-P-1385, with both class 1 primer, and enamel thinned for spraying with thinner conforming to TT-T-1386. Other thinners or primers might be used, however, to accommodate certain specific application requirements.

6.2 Shelf storage life. The following tests are suggested to evaluate suitability for use after prolonged storage.

6.2.1 Class 1. After a 500-ml sample is stored in a tightly sealed glass jar at standard conditions (see 4.4) for one year after date of manufacture, the enamel shall meet the requirements of 3.4.1 and 3.4.5, and the consistency and drying time shall not vary more than plus or minus 20 percent of the values in table II.

6.2.2 Class 2. After the filled pressurized dispensers of the enamels have been stored for one year at standard conditions (see 4.4), the following requirements shall be met:

(a) Spraying properties (see 3.3.2).
(b) Drying time (see table II).
(c) Contrast ratio (see table I).
(d) Color (see 3.4.5).

6.3 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Color required (see 3.4.5).
(d) Level of packaging and level of packing required (see 5.1).
(e) Special marking required (see 5.5).
(f) Size of container required (see 5.1 and 3.3).

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5.3 Palletization. When shipments to Government depots are full car or truckload, the shipping containers shall be palletized on expendable wooden pallets, 2 or 4 way design, to facilitate handling, in accordance with normal commercial practice. The palletized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be as specified in the contract or order (see 6.2).

5.5 DOT regulations. The packaging, packing, marking, and labeling shall comply with 49 CFR 171-178.

5.6 Special marking. Legible commercial marking shall be acceptable. Cans overcoated to obliterate original marking shall not be acceptable. Marking shall be lithographed on the unit container or printed on a lable securely affixed to the container and shall include the following:

(a) Name of material.
(b) Volume in units of US gallons (class 1).
(c) Net weight in ounces (class 2).
(d) Color number of Fed. Std. No. 595.
(e) Date of manufacture, in month and year.
(f) Date of first reinspection (Insert date one year after date of manufacture.)
(g) "Do not puncture or incinerate." (class 2 only).
(h) "Keep away from fire or flame." (class 2 only).
(i) "Use with adequate ventilation."
(j) "Store below 49°C (120"F). (class 2 only).

6. NOTES

6.1 Intended use. This enamel is used where there is need for a relatively fast drying, high gloss coating applied to automotive equipment, postal letter boxes, and other metal surfaces subject to sustained exterior exposure. The conventional ready mixed class 1 enamel may be brushed, but it is suited more to spray applications. The class 2 enamel is intended primarily for touch-up of the class 1 enamel.

6.1.1 Coating system. This enamel is intended as part of a coating system where the surface is first primed with primer coating conforming to TT-P-1385, with both class 1 primer, and enamel thinned for spraying with thinner conforming to TT-T-1366. Other thinners or primers might be used, however, to accommodate certain specific application requirements.

6.2 Shelf storage life. The following tests are suggested to evaluate suitability for use after prolonged storage.

6.2.1 Class 1. After a 500-ml sample is stored in a tightly sealed glass jar at standard conditions (see 4.4) for one year after date of manufacture, the enamel shall meet the requirements of 3.4.1 and 3.4.5, and the consistency and drying time shall not vary more than plus or minus 20 percent of the values in table II.

6.2.2 Class 2. After the filled pressurized dispensers of the enamels have been stored for one year at standard conditions (see 4.4), the following requirements shall be met:

(a) Spraying properties (see 3.3.2).
(b) Drying time (see table II).
(c) Contrast ratio (see table I).
(d) Color (see 3.4.5).

6.3 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Color required (see 3.4.5).
(d) Level of packaging and level of packing required (see 5.1).
(e) Special marking required (see 5.5).
(f) Size of container required (see 5.1 and 3.3).

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6.4 Supersession data. This specification supersedes General Services Administration Purchase Descriptions as follows:

TT-E-001384	GSA Purchase Description	Color in Fed. Std. No. 595
Class 1	PD-FSS-1A	11136
	PD-FSS-2A	17773
	PD-FSS-3A	15050
	PD-FSS-4A	14064
Class 2	PD-FSS-8	11136
	PD-FSS-9	17773
	PD-FSS-10	15050
	PD-FSS-11	14064

GSA DC-01807181

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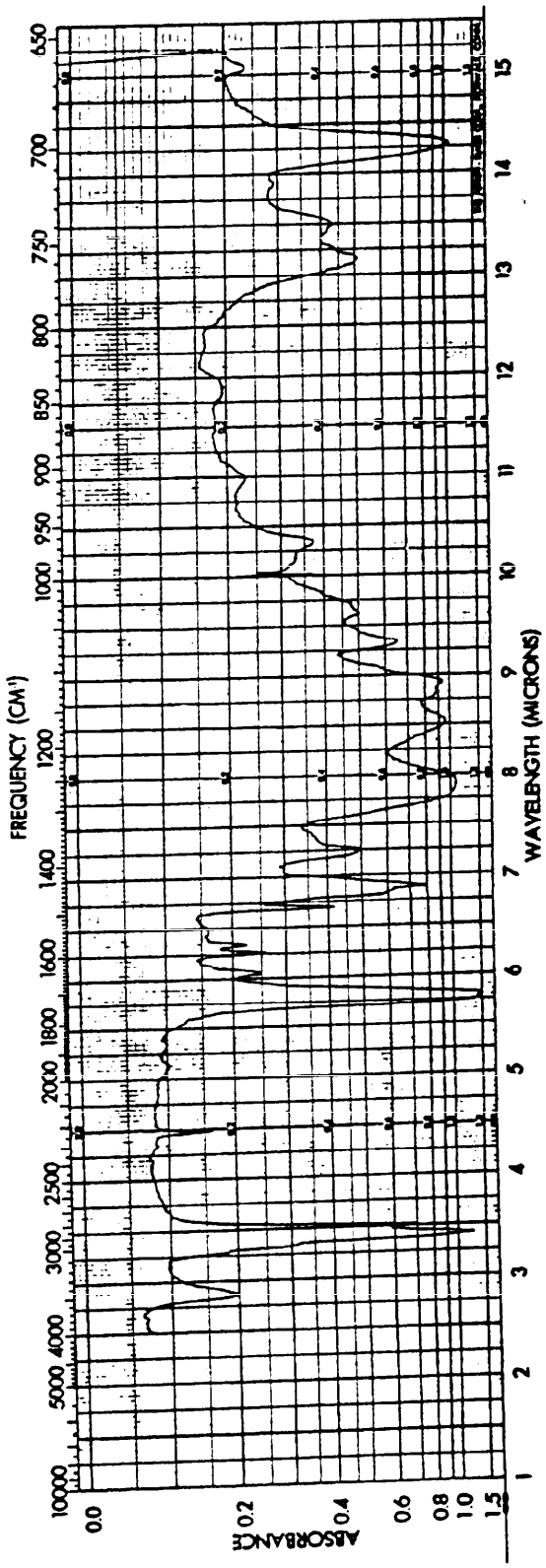


Figure 1. Infrared Analysis of Nonvolatile Vehicle

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